STAT 341 Notes

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1 Introduction

1.1 The Statistical Narrative

We direct attention towards populations,

- the units which define them
- variates as measures on the units

All genuine populations are finite

1.2 Population Attributes

Summaries of finite populations (population attributes)

- Define attributes of interest will rely on statistical concepts
- Population attributes might involve considerable calculation
- No Probability appears for some time as we have the entire population
- Estimate algorithms appear any samples are obtained

1.3 Sample

All possibe samples

- are explored and characteristics of various population attributes examined within this context
- This is done without probability

All possible samples entail a combinatorial explosion for calculation

- the notion of selecting s subset of the samples is introduced
- Probability is used for the selection of a sample froom all possible samples

1.4 Prediction

- Interest lies in predicting the value of a variate (the response variate) given the value of some explanatory variates
- Build a response model that encodes how that prediction is to be carried out
- How should the accuracy be measured?
  - Bias-variance tradeoff
  - training and test and cross-validation

2 Populations

2.1 Populations

A population is a finite (though possibly huge) set $\mathcal{P}$ of elements

- Elements of a population are called unites $u \in \mathcal{P}$
- Variates are functions $x(u), y(u)$, etc. on the which individual unites $u \in \mathcal{P}$. For simplicity we will more often use the notation x_u, y_u, etc. when referring to the realized values of these variates for the unit $u = 1, \cdots, N$

Population attributes, denoted as $a(\mathcal{P})$. In particular, we will

- Define them
- Consider how to calculate them
- Evaluate some of their (non-sampling) properties, e.g. interpretation of the characteristic being captured, sensitivity to outlying points, etc.

2.2 Explicitly Defined Population Attributes

The population is typically a set or collection of units, each with one or more variate that we can measure

2.2.1 Population Attributes

Variates

Variates are characteristics of each unit in the population, and they can take on numerical or categorical values

- The values of variates typically differ from unit to unit
- If we are only interested in the variate y's we might write the population as

$$\mathcal{P} = \{y_1, y_2, \cdots, y_N\}$$

Population Attributes

Population attributes are summaries describing characteristics of the population

- An attribute is a function applied to the entire population and determined by the variate values observed for each of the population's units

$$a(\mathcal{P}) = f(y_1, y_2, \cdots, y_N)$$

- Some examples of attributes are:

- the population total:

$$a(\mathcal{P}) = \sum_{u \in \mathcal{P}} y_u$$

- various counts over the population:

$$a(\mathcal{P}) = \sum_{u \in \mathcal{P}} I_A(y_u)$$

  where $I_A(y)$ is the indicator function

$$I_A(y) = \begin{cases} 1 & \text{if } y \in A \\ 0 & \text{otherwise} \end{cases}$$

Location Attributes

These attributes measure or describe the centre of the distribution of variate values in a dataset

- population average/mean:

$$a(\mathcal{P}) = \overline{y} = \frac{1}{N} \sum_{u \in \mathcal{P}} y_u$$

- population proportion:

$$a(\mathcal{P}) = \frac{1}{N} \sum_{u \in \mathcal{P}} I_A(y_u)$$

- Others include mode, median, etc.

Spread Attributes

These attributes measure variability or spread of the variate values in a data set

- population variance:

$$a(\mathcal{P}) = \frac{1}{N} \sum_{u \in \mathcal{P}} (y_u - \overline{y})^2$$

- population standard deviation:

$$a(\mathcal{P}) = SD_{\mathcal{P}}(y) = \sqrt{\frac{1}{N} \sum_{u \in \mathcal{P}} (y_u - \overline{y})^2}$$

- coefficient of variation:

$$a(\mathcal{P}) = \frac{SD_{\mathcal{P}}(y)}{\overline{y}}$$

- Note: population variance or standard deviation could also be defined using $N - 1$ in the denominator

- Others include range, inter-quartile range, etc.

Order Statistics

Population attributes can also be based on an indexed collection of values

$$y_{(1)} \leq y_{(2)} \leq \cdots \leq y_{(N)}$$

whihch are the variates values $y_u \in \mathcal{P}$ ordered from smallest to largest (including ties)

Location Attributes Based on Order Statistics

- population minimum:
$$a(\mathcal{P}) = \underset{u \in \mathcal{P}}{min} y_u = y_{(1)}$$
- population maximum:
$$a(\mathcal{P}) = \underset{u \in \mathcal{P}}{max} y_u = y_{(N)}$$
- population mid range:
$$a(\mathcal{P}) = \frac{y_{(1)} + y_{(N)}}{2}$$
- population median:
$$a(\mathcal{P}) = \begin{cases} \frac{y_{(N/2)} + y_{(N/2+1)}}{2} & \text{if } N \text{ is even} \\ y_{((N+1)/2)} & \text{if } N \text{ is odd} \end{cases}$$
- population quartiles:
  - Q_1 is 25th percentile, or the first quantile
  - Q_2 is 50th percentile, or the second quantile (median)
  - Q_3 is 75th percentile, or the third quantile

Variability Attributes Based on Order Statistics

- population range:
$$a(\mathcal{P}) = y_{(N)} - y_{(1)}$$
- population inter-quantile range IQR:
$$a(\mathcal{P}) = Q_3 - Q_1$$
- Mdeian Absolute Deviation (MAD) is the median of the absolute differences between eahc y_u and the median
$$a(\mathcal{P}) = \underset{u \in \mathcal{P}}{median} |y_u - \underset{u \in \mathcal{P}}{median} y_u|$$

Skewness Attributes

THese are measures of asymmetry in a population. A symmetric distribution of population values should result in a skewness attribute of zero

- Pearson's moment coefficient of Skewness:

$$a(\mathcal{P}) = \frac{\frac{1}{N}\sum_{u\in\mathcal{P}}(y_u - \overline{y})^3}{[SD_{\mathcal{P}}(y)]^3}$$

- Pearson's second skewness coefficient (median skewness) given by

$$a(\mathcal{P}) = 3 \times \frac{(\overline{y} - \underset{u\in\mathcal{P}}{median} y_u)}{SD_{\mathcal{P}}(y)}$$

- Bowley's measure of skewness based on the quartiles

$$a(\mathcal{P}) = \frac{(Q_3 + Q_1)/2 - Q_2}{(Q_3 - Q_1)/2}$$

NAs in R

Many programs in R accommodate missing data (represented as **NA**s)

- We can choose to simply omit **NA**s, we can use the `na.omit()` function

2.2.2 Attributes Properties

Location Invariance and Equivariance

For an attribute $a(\mathcal{P}) = a(y_1, \cdots, y_N)$ we say that for any $m > 0$ and $b \in \mathbb{R}$, that the attribute is

- location invariant if

$$a(y_1 + b, \cdots, y_N + b) = a(y_1, \cdots, y_N)$$

- location equivariant if

$$a(my_1 + b, \cdots, my_N + b) = ma(y_1, \cdots, y_N) + b$$

Population average is location equivariant

Scale Invariance and Equivariance

For an attribute $a(\mathcal{P}) = a(y_1, \cdots, y_N)$ we say that for any $m > 0$ and $b \in \mathbb{R}$, that the attribute is

- scale invariant if

$$a(m \times y_1, \cdots, m \times y_N) = a(y_1, \cdots, y_N)$$

- scale equivariant if
$$a(m \times y_1, \cdots, m \times y_N) = m \times a(y_1, \cdots, y_N)$$
- location-scale invariant if it is both location invariant and scale invariant
$$a(m \times y_1 + b, \cdots, m \times y_N + b) = a(y_1, \cdots, y_N)$$
- location-scale equivariant if it is both location equivariant and scale equivariant
$$a(m \times y_1 + b, \cdots, m \times y_N + b) = m \times a(y_1, \cdots, y_N) + b$$

Population variance is location-scale equivariant

Replication

The attribute $a(\mathcal{P})$ is

- replication invariant whenever
$$a(\mathcal{P}^k) = a(\mathcal{P})$$
- replication equivariant whenever
$$a(\mathcal{P}^k) = k \times a(\mathcal{P})$$

Population average is replication invariant

2.2.3 Influence, Sensitivity Curves, and Breakdown Points

Influence

If we remove variate y_u then the influence of that variate on the population attribute is quantified by
$$\triangle(a, u) = a(y_1, \cdots, y_{u-1}, y_u, y_{u+1}, \cdots, y_N) - a(y_1, \cdots, y_{u-1}, y_{u+1}, \cdots, y_N)$$
Ideally, no signle unit's value should have greater influence than others

Sensitivity Curves

We can examine the effect on an attribute when we add a variate. To examine the effect,

- suppose we have a population of size $N - 1$
- add a variate with the value y
- new population with N elements is $\{y_1, \cdots, y_{N-1}, y\}$

Define the sensitivity curve of an attribute as

$$SC(y; a(\mathcal{P})) = N[a(y_1, \cdots, y_{N-1}, y) - a(y_1, \cdots, y_{N-1})]$$

General sensitivity curve function in R can be defined as

```
sc = function(y.pop, y, attr, ...) {
    N <- length(y.pop) + 1
    sapply( y, function(y.new) {
        N * attr(c(y.new, y.pop), ...) - N * attr(y.pop, ...)
    })
}
```

Breakdown Points

Gives an assessment of jusst how large a proportion of the data must be contaminated before the statistic breaks down (becomes useless)
The breakdown point of a statistic is the smallest possible fraction of the observations that can be changed to somethings very extreme to make the error large (infinite)
Attributes with high breakdown points are called resistant or robust

2.2.4 Graphical Attributes

Histograms

Helps determine how the values are concentrated
We can define bins in two ways

- bins of equal size
- bins with equal number of elements but varying size

Rules for Number of Bins

- Sturges' rule:

$$\text{Number of bins} = \lceil \log_2(N) + 1 \rceil$$

- Freedman-Diaconis rule:

$$\text{Bin size} = 2 \times \frac{IQR}{N^{1/3}}$$

- Scott's rule:

$$\text{Bin size} = 3.5 \times \frac{SD}{N^{1/3}}$$

Scatter-Plots

Used to see whether two variates x and y are related in some way

2.2.5 Power Transformations

For variates y, sometimes can be helpful to re-express the values in a non-linear way (with transformation $T(y)$) so that they are easier to define, understand, or simply to determine
A commonly used transformation when $y > 0$ is power transformation, which is indexed by a power α. The general form is

$$T_\alpha(y) = \begin{cases} y^\alpha & \alpha \neq 0 \\ \log(y) & \alpha = 0 \end{cases}$$

There transformations are monotonic

$$y_u < y_v \iff T_\alpha(y_u) < T_\alpha(y_v)$$

What changes, often dramatically, is the relative positions of the variate values
The most common purpose of a transformation is to change the shape of the histogram so that it is more symmetric Another power transformation is

$$T_\alpha(y) = \begin{cases} y^\alpha & \alpha > 0 \\ \log(y) & \alpha = 0 \\ -(y)^\alpha & \alpha < 0 \end{cases}$$

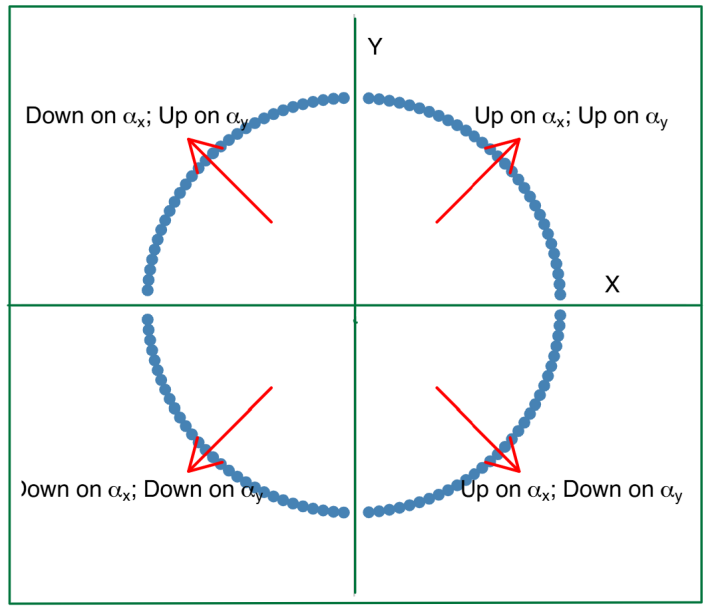
Bump Rule 1: Making Histograms More Symmetric

- If the bump is on "lower" values, then move the power "lower" on the ladder
- If the bump is on "higher" values, then move the power "higher" on the ladder

Bump Rule 2: Straightening Scatter-Plots

Since use of power transformations on both x and y, we will have two different ladders of transformations

- x ladder
- y ladder



2.2.6 Order, Rank, and Quantiles

Recall the order statistics:

$$y_{(1)} \leq y_{(2)} \leq \cdots \leq y_{(N)}$$

which are the ordered values (including ties) of the variate values $y_u \in \mathcal{P}$
The rank statistics (index of y_i)

$$r_1, r_2, \cdots, r_N$$

which are the ranks of the variate values $y_1, y_2, \cdots, y_N$ from the $y_u \in \mathcal{P}$

Quantiles

- $p \in \{\frac{1}{N}, \frac{2}{N}, \cdots, 1\}$ and
- $Q_y(p)$ is the p-th quantile of y

  $$Q_y(p) = y_{(N \times p)}$$

  is called the quantile function of y for all $p \in [\frac{1}{N}, 1]$
- the quantile $Q_y(p)$ for any p locates the variate valuues in the population, and is thus a measure of location
- most (but not all) location measures try to capture central tendency

Quantiles that Measure Center

- the median: $Q_y(1/2)$
- the mid-hinge (average of the first and third quantiles): $\dfrac{Q_y(1/4) + Q_y(3/4)}{2}$
- the mid range (average of the minimum and maximum): $\dfrac{Q_y(1/N) + Q_y(1)}{2}$
- the trimean: $\dfrac{Q_y(1/4) + 2 \times Q_y(1/2) + Q_y(3/4)}{4}$

Quantiles that Measure Spread

The quantile function can be used to provide some natural measures of spread for the variate y:

- the range: $Q_y(1) - Q_y(1/N)$
- the inter-quantile rnage: $IQR_y = Q_y(3/4) - Q_y(1/4)$
- the central $100 \times p\%$ range
- the difference between any two quantiles might be divided by the difference in the corresponding p values, which is the slope

Concentration in Quantiles Plots

We draw boxes of fixed height to see how many elements are within the box $\Rightarrow$ how concentration changes with p

2.3 Implicit Attributes

2.3.1 The Minimum of a Function

We want the value $\hat{\theta}$ which satisfies

$$\hat{\theta} = \underset{\theta}{argmin} \quad \rho(\theta; \mathcal{P})$$

where the possible values of θ may be constrained to be in some set Θ The maximizing function is the same as minimizing the negation, so we only need to consider minimization
The most common form for $\rho(\theta, \mathcal{P})$ is a sum of functions $\rho(\theta, u)$ evaluated at each unit $u \in \mathcal{P}$

$$\rho(\theta, \mathcal{P}) = \sum_{u \in \mathcal{P}} \rho(\theta, u)$$

2.3.2 Dealing with Influential Units in Linear Regression

By applying power transformations to the variates, we can see variates that do not behave as the others do. It is likely that these will have a large influence on the fitted regression line
Thus we can either

- remove the units
- assign weights to the observations according to their variation
- use a method to find the regression line which is robust to potential outliers

Weighted Linear Regression

In weighted linear regression (WLS), the fitted line minimizes the following objective function

$$(\hat{\alpha}, \hat{\beta}) = \underset{\alpha,\beta}{argmin} \quad \sum_{u \in \mathcal{P}} w_u (y_u - \alpha - \beta(x_u - c))^2$$

We assume the weight w_u are known
It is common to either set $c = 0$ or define c to be the weighted average of the x_u values

$$c = \overline{x}_w = \frac{\sum_{u \in \mathcal{P}} w_u x_u}{\sum_{u \in \mathcal{P}} w_u}$$

We estimate α and β as

$$\hat{\alpha} = \overline{y}_w - \hat{\beta}(\overline{x}_w - c)$$

$$\hat{\beta} = \frac{\sum_{u \in \mathcal{P}} w_u (x_u - \overline{x}_w)(y_u - \overline{y}_w)}{\sum_{u \in \mathcal{P}} w_u (x_u - \overline{x}_w)^2}$$

where $\overline{y}_w$ and $\overline{x}_w$ are the weighted averages of y_u and x_u respectively

Robust Regression

Points that are far from the linear line should have less weight in the objective function. The objective function will be

$$(\hat{\alpha}, \hat{\beta}) = \underset{\alpha,\beta}{argmin} \quad \sum_{u \in \mathcal{P}} \rho(y_u - \alpha - \beta(x_u - c))$$

In robust regression we modify the loss function $\rho(y_u - \alpha - \beta(x_u - c))$ so that

- gives lower weight than least squares to units with large residuals
- it is quadratice near 0 and hence behaves similarly to LS for units with small residuals

Huber Loss Function

It achieves these goals by combining the quadratice and absolute value functions

$$\rho_k(r) = \begin{cases} \frac{1}{2}r^2 & |r| < k \\ k(|r| - k/2) & |r| \geq k \end{cases}$$

Another form of robust regressino involves defining the loss function in terms of least absolute deviations (LAD)

$$(\hat{\alpha}, \hat{\beta}) = \underset{\alpha,\beta}{argmin} \quad \sum_{u \in \mathcal{P}} |y_u - \alpha - \beta(x_u - c)|$$

2.3.3 Gradient Descent

Direction and Step Size

If $\rho(\theta; \mathcal{P})$ is a differentiable function of $\theta = (\theta_1, \cdots, \theta_p) \in \mathbb{R}^k$, then we can calculate the gradient for any value of θ

$$g = g(\theta) = \triangle\rho(\theta; \mathcal{P}) = \begin{bmatrix} \dfrac{\partial\rho(\theta; \mathcal{P})}{\partial\theta_1} \\ \vdots \\ \dfrac{\partial\rho(\theta; \mathcal{P})}{\partial\theta_k} \end{bmatrix}$$

By definition, the normalized gradient

$$d_i = \frac{g_i}{||g_i||}$$

provides the direction in which $\rho(\theta; \mathcal{P})$ increases or decreases fastest

- d_i indicates the direction of steepest ascent
- $-d_i$ incidates the directino of steepest descent

We iterate and obtain a new estimate of θ by

- moving in the direction of $-d_i$ and
- taking a step of size $\lambda_i > 0$

$$\hat{\theta}_{i+1} = \hat{\theta}_i - \lambda_i d_i$$

Note that the step size λ at each iteration can been chosen in a variety of ways

1. could choose a fied value for all i such as $\lambda_i = 0.1$
2. could define a fixed sequence such as $\lambda_i = 0.1 + 1/i$
3. could perform a line search and algorithmically choose the value of λ_i that minimizes

$$\rho(\hat{\theta}_i - \lambda_i d_i)$$

The Gradient Descent Algorithm

Given some initial values $\hat{\theta}_0$

1. initialize $i = 0$
2. Loop until convergence:
   a. Gradient:
   $$g_i = \nabla \rho(\hat{\theta}_i; \mathcal{P})$$
   b. Gradient direction:
   $$d_i = \frac{g_i}{||g_i||}$$
   c. Line search: Find the step size $\hat{\lambda}_i$
   $$\hat{\lambda}_i = \underset{\lambda > 0}{argmin} \quad \rho(\hat{\theta}_i - \lambda d_i)$$
   d. Update iterate:
   $$\hat{\theta}_{i+1} = \hat{\theta}_i - \hat{\lambda}_i d_i$$
   e. Converged?
   - if iteratesnot changing, Return
   - else $i = i + 1$ and repeat
3. Return $\hat{\theta} = \hat{\theta}_i$

R Code for Gradient Descent

```
gradientDescent <- function(theta = 0, rhoFn, gradientFn, lineSearchFn,
    testConvergenceFn, maxIterations = 100, tolerance = 1e-06, relative = FALSE,
    lambdaStepsize = 0.01, lambdaMax = 0.5) {

    converged <- FALSE
    i <- 0

    while (!converged & i <= maxIterations) {
    g <- gradientFn(theta)       ## gradient
    glength <- sqrt(sum(g^2))    ## gradient direction
    if (glength > 0)
        d <- g / glength

    lambda <- lineSearchFn(theta, rhoFn, d, lambdaStepsize = lambdaStepsize,
                           lambdaMax = lambdaMax)

    thetaNew <- theta - lambda * d
    converged <- testConvergenceFn(thetaNew, theta, tolerance = tolerance,
                                   relative = relative)
    theta <- thetaNew
    i <- i + 1
    }

    ## Return last value and whether converged or not
    list(theta = theta, converged = converged, iteration = i, fnValue = rhoFn(
        theta))
}
```

2.3.4 Systems of Equations

Scalar Valued Attributes

The average is the value of θ that solves

$$\psi(\theta; \mathcal{P}) = \sum_{u \in \mathcal{P}} \psi(\theta; u) = \sum_{u \in \mathcal{P}} y_u - n\theta = 0$$

The weighted average is the value of θ which solves

$$\psi(\theta; \mathcal{P}) = \sum_{u \in \mathcal{P}} \psi(\theta; u) = \sum_{u \in \mathcal{P}} w_u(y_u - \theta) = 0$$

The q-th quantile is the samllest θ which solves

$$\psi(\theta; \mathcal{P}) = \sum_{u \in \mathcal{P}} \psi(\theta; u) = \sum_{u \in \mathcal{P}} \frac{1}{N} I(y_u \leq \theta) - q = 0$$

Vector Valued Attributes

Least squares

$$\psi(\theta;\mathcal{P}) = \sum_{u\in\mathcal{P}} \psi(\theta;u) = \sum_{u\in\mathcal{P}} (y_u - \alpha - \beta(x_u - c)) \begin{bmatrix} 1 \\ x_u - c \end{bmatrix} = 0$$

Weighted least squares

$$\psi(\theta;\mathcal{P}) = \sum_{u\in\mathcal{P}} \psi(\theta;u) = \sum_{u\in\mathcal{P}} w_u(y_u - \alpha - \beta(x_u - c)) \begin{bmatrix} 1 \\ x_u - c \end{bmatrix} = 0$$

Robust regression

$$\psi(\theta;\mathcal{P}) = \sum_{u\in\mathcal{P}} \psi(\theta;u) = \sum_{u\in\mathcal{P}} \rho_k'(y_u - \alpha - \beta(x_u - c)) \begin{bmatrix} 1 \\ x_u - c \end{bmatrix} = 0$$

2.3.5 Newton's Method

Suppose we have a differentiable function $f(x)$ and we wish to find $x = x^*$ which solves

$$f(x) = 0$$

Given initial value x_0

- use a linear function to approximate $f(x)$ in the vicinity of x_0

$$f(x) \approx f(x_0) + f'(x_0)(x - x_0)$$

- then find the root of linear approximation to iterate to the next value of x

$$0 = f(x_0) + f'(x_0)(x - x_0) \implies x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

- continue until x_{i+1} does not differ much from x_i

For implicitly defined scalar attribute θ, we want to find $\theta \in \Theta$ such that

$$\psi(\theta;\mathcal{P}) = 0$$

- Given current guess $\hat{\theta} = \hat{\theta}_i$ a first order approximation is

$$\psi(\theta;\mathcal{P}) \approx \psi(\hat{\theta}_i;\mathcal{P}) + \psi'(\hat{\theta}_i;\mathcal{P}) \times (\theta - \hat{\theta}_i)$$

- Then the update is the root of the linear approximation and is given by

$$\hat{\theta}_{i+1} = \hat{\theta}_i - \frac{\psi(\hat{\theta}_i;\mathcal{P})}{\psi'(\hat{\theta}_i;\mathcal{P})}$$

The Newton Algorithm

Given an initial value $\hat{\theta}_0$

1. Initialize: $i = 0$
2. Loop:
   (a) Update iterate:
   $$\hat{\theta}_{i+1} = \hat{\theta}_i - \frac{\psi(\hat{\theta}_i; \mathcal{P})}{\psi'(\hat{\theta}_i; \mathcal{P})}$$
   (b) Converged?
   - if iterates not changing, return
   - else $i = i + 1$ and repeat
3. Return $\hat{\theta} = \hat{\theta}_i$

2.3.6 The Newton raphson Algorithm

The multivariate analog of Newton's method
Given an initial value $\hat{\theta}_0$

1. Initialize: $i = 0$
2. Loop:
   (a) Update iterate:
   $$\hat{\theta}_{i+1} = \hat{\theta}_i - [\psi'(\hat{\theta}_i; \mathcal{P})]^{-1} \psi(\hat{\theta}_i; \mathcal{P})$$
   (b) Converged?
   - if iterates not changing, return
   - else $i = i + 1$ and repeat
3. Return $\hat{\theta} = \hat{\theta}_i$

2.3.7 Iteratively Reweighted Least Squares (IRLS)

Given initial value $\hat{\theta}_0$

1. Initialize: $i = 0$
2. Loop:
   (a) Construct residuals and weights for all $u \in \mathcal{P}$
   $$r_u = y_u - z_u' \theta_i \quad w_u = \frac{\rho'(r_u)}{r_u}$$
   where $z_u = \begin{bmatrix} 1 & x_u - c \end{bmatrix}'$

(b) Solve the weighted least squares problem

$$\sum_{u \in \mathcal{P}} w_u (y_u - z_u' \theta) z_u = 0$$

(c) Update parameter:

$$\hat{\theta}_{i+1} = \hat{\theta}_i$$

(d) Converged?

- if iterates not changing, return
- else $i = i + 1$ and repeat

3. return $\hat{\theta} = \hat{\theta}_i$

3 Samples

If we have a sample or a subset S of $n << N$ units, then the attribute $a(S)$ calculated based on this sample is an estimate of its population counterpart $a(\mathcal{P})$

$$a(S) = a(\hat{\mathcal{P}}) = a(\hat{\mathcal{P}})$$

Sample Error

Any difference between the actual values of the estimates $a(S)$ and the quantity being estimated (the estimand) $a(\mathcal{P})$ is an error

$$\text{sample error} = a(S) - a(\mathcal{P})$$

The nature of this error will depend on the sample and attribute

Fisher Consistency

If the sample S is equal to the population $\mathcal{P}$ then the sample error should be zero (or non-existent), $a(S) = a(\mathcal{P})$
This would mean the estimation is in some sense consistent, called Fisher consistency

3.1 All Possible Samples

Sample Error

The sample error for a sample S of size n is

$$a(S) - a(\mathcal{P}) = \frac{1}{n}\sum_{u \in S} y_u - \frac{1}{N}\sum_{u \in \mathcal{P}} y_u$$

Average Sample Error

The average sample error over all possible samples of size n is

$$\text{Average Sample Error} = (\frac{1}{M}\sum_{i=1}^{M} a(S_i)) - a(\mathcal{P})$$

3.1.1 Consistency and the Effect of Sample Size

The nature of sample error depends largely on the sample size

- As sample size increases, the sample approaches the population
- Attribute values will concentrate even more around the population value

We can quantify the concentration by

$$||a(S) - a(\mathcal{P})|| = ||\frac{1}{n}\sum_{u \in S} y_u - \frac{1}{N}\sum_{u \in \mathcal{P}} y_u|| < c$$

for some $c > 0$, we could proportion of samples that satisfy this
Consider a population $\mathcal{P}$ of size $N < \infty$

- For each n, construct the set of all possible samples

$$\mathcal{P}_S(n) = \{S : S \subset \mathcal{P}, |S| = n\}$$

- For any $c > 0$,

$$\mathcal{P}_a(c, n) = \{S : S \subset \mathcal{P}_S(n), ||a(S) - a(\mathcal{P})|| < c\}$$

  and define the proportion

$$p_a(c, n) = \frac{|\mathcal{P}_a(c, n)|}{|\mathcal{P}_S(n)|}$$

  for all $c > 0$ and $n \leq N$

3.1.2 Comparisons Across Attributes

We use relative absolute sample error to compare different attributes

$$\mathcal{P}_a^*(c, n) = \{S : S \subset \mathcal{P}_S(n), \frac{||a(S) - a(\mathcal{P})||}{||a(\mathcal{P})||} < c\}$$

and define the corresponding proportino, for all $c > 0$ and $n \leq N$

$$p_a^*(c, n) = \frac{|\mathcal{P}_a^*(c, n)|}{|\mathcal{P}_S(n)|}$$

3.2 Selecting Samples

3.2.1 Randomly Selecting m Samples

Consider drawing samples of size n from population $\mathcal{P}$. The population $\mathcal{P}_S$ of all such samples has size $M = \binom{N}{n}$ and is denoted

$$\mathcal{P}_S = \{S_1, S_2, \cdots, S_M\}$$

Any attribute $a(S_u)$ is now just a variate on that unit
We have a population of attributes

$$\mathcal{P}_{a(S)} = \{a(S_1), a(S_2), \cdots, a(S_M)\}$$

3.2.2 Quantifying Sample Error

In principle we select a sample S from the population $\mathcal{P}_S$ contianing all possible samples

- We do so with some probability $p(S) \geq 0$ of being selected, require

$$\sum_{S \in \mathcal{P}_S} p(S) = 1$$

- For any sample $S \in \mathcal{P}_S$, we have sample error

$$\text{Sample Error} = a(S) - a(\mathcal{P})$$

- Aveerage sample error is

$$\text{Average Sample Error} = \frac{1}{M} \sum_{S \in \mathcal{P}_S} [a(S) - a(\mathcal{P})]$$

3.2.3 Sampling Bias, Sampling Variance, and Sampling Mean Squared Error

Sampling Bias

$$\begin{aligned} \text{Sampling Bias} &= E[a(S)] - a(\mathcal{P}) \\ &= \sum_{S \in \mathcal{P}_S} a(S)p(S) - a(\mathcal{P}) \\ &= \sum_{S \in \mathcal{P}_S} [a(S) - a(\mathcal{P})]p(S) \end{aligned}$$

Sampling bias is the expected sample error induced by the repeated random sampling of S from $\mathcal{P}_S$
If sampling bias is zero, then $a(S)$ is unbiased estimator of $a(\mathcal{P})$

Sampling Variance

$$Var[a(S)] = E[(a(S) - E(a(S)))^2]$$

This quantifies dispersion in sample errors

Mean Squared Error

Use to quantify the expected squared distance between $a(S)$ and $a(\mathcal{P})$

$$MSE[a(S)] = E[(a(S) - a(\mathcal{P}))^2] = Var[a(S)] + [\text{Sampling Bias}]^2$$

3.2.4 Sampling Mechanisms

A sequence of the first k units u_i selected from $\mathcal{P}$ is

$$s_k = (u_{i1}, u_{i2}, \cdots, u_{ik})$$

A sampling mechanism is defined by the probabilities

$$Pr(u) \text{ and } Pr(u|k, s_{k-1})$$

where

- the first unit is selected with probabilityy $Pr(u)$ and
- the probability of the sequence of the first k units selected is

$$Pr(s_k) = Pr(u_{i1})Pr(u_{i2}|2, s_1) \cdots Pr(u_{ik}|k, s_{k-1})$$

Simple Random Sampling Without Replacement (SRSWOR)

The sampling mechanism is

$$Pr(u) = \frac{1}{N} \text{ and } Pr(u|k, s_{k-1}) = \frac{1}{N-k+1}$$

The probability of the sequence s_n is

$$Pr(s_n) = \frac{1}{N}\frac{1}{N-1}\cdots\frac{1}{N-n+1}$$

which is the same for all $n!$ permutations, so

$$p(S) = \frac{n!}{N(N-1)\cdots(N-n+1)} = \frac{1}{\binom{N}{n}}$$

Simple Random Sampling With Replacement (SRSWR)

The sampling mechanism is

$$Pr(u) = \frac{1}{N} = Pr(u|k, s_{k-1})$$

Sample S can contain one or more replicated units
The probability of the sequence s_n is

$$Pr(s_n) = \frac{1}{N^n}$$

3.2.5 Unit Inclusion Probabilities

The inclusion probability for unit u is probability of unit u being included in the sample

$$\pi_u = Pr(u \in S) = \sum_{S \in \mathcal{P}_S} p(S) \times I(u \in S)$$

Such inclusion probabilities are of interest to calculate in addition to $p(S)$ and may in fact be derived from $p(S)$
Consider the following random variable

$$D_u = \begin{cases} 1 & \text{if } u \in S \\ 0 & \text{if } u \notin S \end{cases}$$

The expectation of this random variable is

$$\begin{aligned} E[D_u] &= 1 \times Pr(D_u = 1) + 0 \times Pr(D_u = 0) \\ &= Pr(u \in S) \\ &= \pi_u \end{aligned}$$

The variance of this random variable is

$$\begin{aligned} Var[D_u] &= E[D_u^2] - E[D_u]^2 \\ &= 1^2 \times Pr(D_u = 1) + 0^2 \times Pr(D_u = 0) - \pi_u^2 \\ &= \pi_u - \pi_u^2 \\ &= \pi_u(1 - \pi_u) \end{aligned}$$

The probability that u and v are in the sample S is called the joint inclusion probability and is given by

$$\pi_{uv} = Pr(u \in S, v \in S)$$

The relationship is

$$\begin{aligned} Cov(D_u, D_v) &= E[D_u \times D_v] - E[D_u]E[D_v] \\ &= Pr(u \in S, v \in S) - \pi_u \pi_v \\ &= \pi_{uv} - \pi_u \pi_v \end{aligned}$$

Simple Random Sampling Without Raplacement (SRSWOR)

The inclusion probability is

$$\pi_u = Pr(u \in S) = \frac{n}{N}$$

The joint inclusion probability is

$$\pi_{uv} = Pr(u \in S, v \in S) = \frac{n(n-1)}{N(N-1)}$$

Simple Random Sampling Withou Replacement (SRSWR)

The inclusion probability is

$$\pi_u = 1 - \left(\frac{N-1}{N}\right)^n$$

The joint probability is

$$\pi_{uv} = 1 - 2\left(\frac{N-1}{N}\right)^n + \left(\frac{N-2}{N}\right)^n$$

3.2.6 Estimating Tools

A natural estimate of a population total

$$a(\mathcal{P}) = \sum_{u \in \mathcal{P}} y_u$$

is the Horvitz-Thompson estimate defined as

$$\hat{a}(\mathcal{P}) = a_{HT}(S) = \sum_{u \in S} \frac{y_u}{\pi_u}$$

where the contribution for each unit in the sample is weighted inversely by π_u, its probability of inclusion in S

- if the probability of inclusion is small, then the weight will be high
- if the probability of inclusion is large, then the weight will be low

The Horvitz-Thompson Estimator

Bias, Variance, and Mean Squared Error

The bias of the HT estimator

$$E[\tilde{a}_{HT}(S) - a(\mathcal{P})] = 0$$

The variance of the HT estimator is

$$Var[\tilde{a}_{HT}(S)] = \sum_{u \in P} \sum_{v \in P} (\pi_{uv} - \pi_u \pi_v) \frac{y_u}{\pi_u} \frac{y_v}{\pi_v} = \sum_{u \in S} \sum_{v \in S} Cov(D_u, D_v) \frac{y_u}{\pi_u} \frac{y_v}{\pi_v}$$

The HT estimator is unbiased, the mean squared error is equal to the variance

$$MSE = Var$$

The HT estimate of the variance of the HT estimator is

$$\hat{Var}[\tilde{a}_{HT}(S)] = \sum_{u \in S} \sum_{v \in S} \left(\frac{\pi_{uv} - \pi_u \pi_v}{\pi_{uv}}\right) \frac{y_u}{\pi_u} \frac{y_v}{\pi_v}$$

Using HT estimation we are able to construct

- an estimate of population total
- an estimate of the variance of this estimator
- both estimators are unbiased

3.2.7 Sampling Designs

The pair $(\mathcal{P}_S, p(S))$ is called a sampling design

- Together they determine which samples are possible and with what probability they are selected
- The SRSWOR, SRSWR and SRSWH frameworks provide examples of different sampling designs
- The sampling design is ours to choose
  - We may choose $\mathcal{P}_S$ so that the values $a(S)$ for $S \in \mathcal{P}_S$ are constrained to be near $a(\mathcal{P})$
  - We may choose $p(S)$ so that samples $S \in \mathcal{P}_S$ that have $a(S)$ close to $a(\mathcal{P})$ have higher probability $p(S)$ of being selected

4 Inference

4.1 Sources of Error

4.1.1 Target and Study Populations

The target population is the population about which we would like to draw an inference
The study population is the population from which samples are taken
The difference between attribute evaluated on the two populations is the **study error**

$$Study\ Error = a(\mathcal{P}_{study}) - a(\mathcal{P}_{target})$$

Then if we obtain a sample S to draw inferences about the target popiulation $\mathcal{P}_{target}$, the error for a given attribute $a(\cdot)$ is

$$\begin{aligned} a(S) - a(\mathcal{P}_{target}) &= [a(S) - a(\mathcal{P}_{study})] + [a(\mathcal{P}_{study}) - a(\mathcal{P}_{target})] \\ &= Sample\ Error + Study\ Error \end{aligned}$$

Future Realizations

The units which are not available at the time of study

4.1.2 Measurement Error

Every measuring system has at least three sources of potential error:

- the measuring device (gauge)
- the person reading or recording the measurement (operator)
- the method followed to take the measurement

4.2 Comparing Sub-Populations

4.2.1 Anatomy of a Significance Test

We would like to quantify, numerically, how unusual the difference between $a(\mathcal{P}_1)$ and $a(\mathcal{P}_2)$ is relative to randomly mixed sub-populations
Following steps are used to gather such evidence:

1. Suppose the sub-populations were randomly drawn from the same population, known as the **null hypothesis**
2. Construct a **discrepency measure** that quantifies how inconsistent our data is with the null hypothesis, large value indicate evidence against the null hypothesis
3. Obtain the **observed discrepency** by calculating the discrepency measure on the observed / unshuffled sub-populations

4. Obtain the **observed p-value** by calculating the probability that a randomly shuffled sub-population has a discrepency measure at least as large as the observed discrepency, small values indicate evidence against the null hypothesis

Null Hypothesis

Each of the following (equivalent) statements constitutes the null hypothesis we are testing:

- H_0: the sub-populations $\mathcal{P}_1$ and $\mathcal{P}_2$ were randomly drawn from the same population
- H_0: $\mathcal{P}_1$ and $\mathcal{P}_2$ were created by randomly assigning units in the same population to one of the two sub-populations
- H_0: $\mathcal{P}_1$ and $\mathcal{P}_2$ were generated by random mixing

Discrepency Measure

A discrepency measure $D(\mathcal{P}_1, \mathcal{P}_2)$ quantifies how inconsistent data is with the null hypothesis, is defined so that large values indicate evidence against the null hypothesis
Form of $D(\mathcal{P}_1, \mathcal{P}_2)$ depends on how we want to compare $\mathcal{P}_1$ and $\mathcal{P}_2$

- compare measures of location, typically based on difference

$$a(\mathcal{P}_1) - a(\mathcal{P}_2)$$

- compare measures of spread, typically based on ratios

$$\frac{a(\mathcal{P}_1)}{a(\mathcal{P}_2)}$$

Observed Discrepency

The observed discrepency d_{obs}, is the value of discrepency measure D calculated on the two observed / unshuffled sub-populations

$$d_{obs} = D(\mathcal{P}_1, \mathcal{P}_2)$$

Observed p-value

The observed p-value is the probability that a randomly shuffled sub-population has a discrepency measure at least as large as the observed discrepency

$$p - value = Pr(D \geq d_{obs} | H_0 \text{ is true})$$

- $p - value < 0.001$: very strong evidence against H_0
- $0.001 < p - value < 0.01$: strong evidence against H_0
- $0.01 < p - value < 0.05$: moderate evidence against H_0

- $0.05 < p - value < 0.1$: weak evidence against H_0
- $p - value > 0.1$: no evidence against H_0
- in extreme cases, $p - value = 0$, the hypothesis must be false

There are $\binom{N_1+N_2}{N_1} = \binom{N_1+N_2}{N_2}$ possible permutations of the observed data
Because that's too many permutations to consider, we typically just use M of them, the p-value is approximated as

$$\frac{1}{M}\sum_{i=1}^{M} I(D(\mathcal{P}_{1,i}, \mathcal{P}_{2,i}) \geq d_{obs})$$

Putting all Together

Test of Significance Algorithm

1. State the null hypothesis: H_0: $\mathcal{P}_1$ and $\mathcal{P}_2$ are drawn from the same population
2. Construct a measure of discrepency $D = D(\mathcal{P}_1, \mathcal{P}_2)$ where large values indicate evidence against the null hypothesis
3. Calculate the observed discrepency $d_{obs} = D(\mathcal{P}_1, \mathcal{P}_2)$
4. Shuffle the sub-populations M times and calculate the observed p-value

$$p - value = Pr(D \geq d_{obs} | H_0 \text{ is true}) \approx \frac{1}{M}\sum_{i=1}^{M} I(D(\mathcal{P}_{1,i}, \mathcal{P}_{2,i}) \geq d_{obs})$$

4.2.2 A t-like Discrepency Measure

When comparing two sub-populations on the basis of a measure of location, one particularly useful discrepancy measure is

$$D(\mathcal{P}_1, \mathcal{P}_2) = \frac{a(\mathcal{P}_1) - a(\mathcal{P}_2)}{\sqrt{Var(\tilde{a}(\mathcal{P}_1)) + Var(\tilde{a}(\mathcal{P}_2))}}$$

This is scale-invariant

4.2.3 Multiple Testing

Estimating d^*_{obs}

- Suppose we have K discrepancy measures $D_1, D_2, \cdots, D_K$
  - The combined discrepancy measure is

$$D^* = 1 - p - value_{min}$$

- For $i = 1, \cdots, M_{inner}$ and each discrepancy $k = 1, \cdots, K$

- randomly mix the two sub-populations $\mathcal{P}_1$ and $\mathcal{P}_2$ yielding $\mathcal{P}^*_{1,i}$ and $\mathcal{P}^*_{2,i}$
- calculate

$$d_{k,i} = D_k(\mathcal{P}^*_{1,i}, \mathcal{P}^*_{2,i})$$

- we estimate each p-value labelled as p_k with

$$\hat{p}_k = \frac{1}{M_{inner}} \sum_{i=1}^{M_{inner}} I(d_{k,i} \geq d_{obs,k})$$

- we estimate $p - value_{min}$ as

$$\hat{d}^*_{obs} = 1 - \min_{k=1,\cdots,K} \hat{p}_k$$

Estimating $p - value^*$

$$p - \hat{v}alue^* = \frac{1}{M_{outer}} \sum_{i=1}^{M_{outer}} I(D^*(\mathcal{P}^*_{1,i}, \mathcal{P}^*_{2,i}) \geq d^*_{obs})$$

4.3 Interval Estimation

4.3.1 Random vs. Observed Intervals

Suppose the attribute of interest is the population average $a(\mathcal{P}) = \mu = \overline{y}$

$$\overline{Y} \sim N(\mu, \frac{\sigma^2}{n}(\frac{N-n}{N-1}))$$

where

$$\sigma^2 = \frac{1}{N} \sum_{u \in \mathcal{P}} (y_u - \mu)^2$$

Random Intervals

using the standardized random variable and sepcified $p \in (0, 1)$ we can find a constant $c > 0$ such that

$$1 - p = Pr(\overline{Y} - c \cdot \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}} \leq \mu \leq \overline{Y} + c \cdot \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}})$$

Observed Intervals

$$\left[\overline{y} - c \cdot \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}} \leq \mu \leq \overline{y} + c \cdot \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}} \right]$$

where $\overline{y}$ is the sample mean and σ is the sample standard deviation

4.3.2 Student t Based Intervals

Standard Error vs. Standard Deviation

The standard error is an estimate of the standard deviation of the corresponding estimator

$$SE[\tilde{a}(\mathcal{P})] = \hat{SD}[\tilde{a}(\mathcal{P})]$$

$$\frac{\overline{Y} - \mu}{\frac{\tilde{\sigma}}{\sqrt{n}}\sqrt{\frac{N-n}{N-1}}} \sim t_{n-1}$$

In the special case that $a(\mathcal{P}) = \overline{Y}$ is the population average then the standard deviation is

$$SD[\tilde{a}(S)] = \frac{\sigma}{\sqrt{n}}\sqrt{\frac{N-n}{N-1}}$$

the standard deviation estimator is

$$\tilde{SD}[\tilde{a}(S)] = \frac{\tilde{\sigma}}{\sqrt{n}}\sqrt{\frac{N-n}{N-1}}$$

the standard error is

$$\hat{SE}[\tilde{a}(S)] = \frac{\hat{\sigma}}{\sqrt{n}}\sqrt{\frac{N-n}{N-1}}$$

4.4 Resampling

4.4.1 Resampling Intuitions

In practice, the population from which our sample was taken cannot generally be repeatedly sampled, we only have one sample
Resampling methods aim to mimic this process by repeatedly sampling S as if it were $\mathcal{P}$
Bootstrapping is a resampling method that sample with replacement from the sample S to create a new sample S^* of the same size as S

4.4.2 Bootstrap

The bootstrap esetimate is, at best, as good as $a(S)$

$$\textit{bootstrap sample error} = a(S^*) - a(\mathcal{S})$$

Bootstrap Standard Deviation

The standard deviation of the corresponding sample attribute's estimator can be estimated from the bootstrap distribution with

$$\hat{SD}_*[\tilde{a}(S^*)] = \sqrt{\frac{\sum_{b=1}^{B}(a(S_b^*) - \overline{a}^*)^2}{B-1}}$$

4.4.3 Bootstrap Confidence Intervals

A 95% naive normal theory bootstrap confidence interval for a population attribute $a(\mathcal{P})$ is

$$a(S) \pm 1.96 \cdot \hat{SD}_*[\tilde{a}(S^*)]$$

4.4.4 Bootstrap t Confidence Intervals

5 Prediction

5.1 Accuracy of Prediction

Prediction

The response model

$$y = \mu(x) + error$$

yields the predictor function $\hat{\mu}(x)$

Accuracy of Prediction

One measure of inaccuracy over the population $\mathcal{P}$ (of size N) is the average prediction squared error (APSE)

$$\frac{1}{N}\sum_{u\in\mathcal{P}}(y_u - \hat{\mu}(x_u))^2$$

5.2 Predictions over Multiple Samples

5.2.1 Calculating APSE Over Many Samples

The inaccuracy of a predictor function $\hat{\mu}(x)$ is measured by its average prediction squared error

$$APSE(\mathcal{P}, \hat{\mu}_S) = \frac{1}{N}\sum_{u\in\mathcal{P}}(y_u - \hat{\mu}_S(x_u))^2$$

The average APSE over all M samples should be a better measure of the quality of a predictor functino

$$APSE(\mathcal{P}, \tilde{\mu}) = \frac{1}{M}\sum_{i=1}^{M} APSE(\mathcal{P}, \hat{\mu}_{S_i}) = \frac{1}{M}\sum_{i=1}^{M}\frac{1}{N}\sum_{u\in\mathcal{P}}(y_u - \hat{\mu}_{S_i}(x_u))^2$$

5.2.2 Decompose $APSE(\mathcal{P}, \tilde{\mu})$

$$\begin{aligned} APSE(\mathcal{P}, \tilde{\mu}) &= \frac{1}{N}\sum_{u\in\mathcal{P}}(y_u - \tau(x_u))^2 + \frac{1}{M}\sum_{j=1}^{M}\frac{1}{N}\sum_{u\in\mathcal{P}}(\hat{\mu}_{S_j}(x_u) - \overline{\hat{\mu}}(x_u))^2 + \frac{1}{N}\sum_{u\in\mathcal{P}}(\overline{\hat{\mu}}(x_u) - \tau(x_u))^2 \\ &= Ave_X(Var[Y|X]) + Var[\tilde{\mu}] + Bias^2[\tilde{\mu}] \end{aligned}$$

5.3 Back to Reality